# A Classifier-Agnostic Bayes-Risk Benchmark for Weakly-Paired Multimodal Fusion, with AVID: A Severely Weakly-Aligned RGB-Infrared Drone Dataset

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**Abstract**

Every existing RGB-thermal fusion benchmark (LLVIP, M3FD, KAIST) is built on co-located or heavily calibrated rigs, so its evaluation protocol silently assumes what classical multi-view learning theory [2] requires: that the two modalities are exact, registered views of the same instance. Real dual-optic sensor rigs violate this, and every existing empirical fix reports the error of *one trained model* – confounded by architecture, training budget, and optimization luck. We instead introduce a **classifier-agnostic Bayes-risk benchmark**: closed-form Bhattacharyya bounds [1, 7] on the *optimal* classifier's error, computed directly from class-conditional feature distributions with no gradient descent anywhere, cross-checked against a nonparametric kernel-density estimate and validated on synthetic data with known separation before being trusted on real patches. Applied to 35,952 real bird/drone instances from AVID (Aerial Visible-Infrared Drone dataset, a large-scale RGB-IR video dataset for counter-drone surveillance), this reveals a fusion advantage 3-4$\times$ larger than any trained model showed us (Bayes bound at the corpus's modal 12-16px scale: RGB=0.148, IR=0.198, **fusion=0.046**); an asymmetric corruption ceiling (RGB's bound roughly triples under noise-type corruptions, IR's barely moves); and a new **Bayes Fusion Gain** metric that shrinks 16% (0.111$\rightarrow$0.093) as injected cross-modal misalignment reaches the real measured ceiling of this corpus's own misalignment (17px, from a real 300-pair sample where 32.3% of pairs exceed 5px against targets that are themselves only 8-16px wide) – instantiated alongside a numerical data-processing-inequality argument showing real mutual information between the fused representation and the label degrading by 9% over the same range. We also introduce the **Alignment Robustness Score (ARS)**, a complementary, architecture-agnostic metric for a single trained model's empirical sensitivity to misalignment (structurally analogous to certified adversarial robustness [4]), and validate it on real YOLOv8n baselines: fusion beats both single modalities (0.685 vs. 0.570 RGB-only / 0.504 IR-only, matched mAP50), with a measured critical radius of 15px – consistent with, and now explained by, the Bayes-risk framework's own findings. Beyond these two benchmarks, we release: 28,790 training / 3,250 validation RGB-IR pairs across 115 videos (after identifying and removing a systematic noise-augmented duplicate the source export had silently baked into every frame); AVID-C, a reproducible ImageNet-C-style corruption suite (7 families $\times$ 5 severities); a measured split-leakage pitfall (naive frame-level splitting leaks 87.5% of "training" frames vs. 0.0% for the video-level splitting we use); and a documented, already-executed cross-modal translation baseline (CycleGAN-Turbo). We also disclose a real data-organization failure found during release preparation – the source Test split's RGB and IR files use non-overlapping numbering schemes with no manifest to resolve them – as a limitation rather than working around it silently. Code (including everything that produced every number above) and the dataset (CC-BY-NC-4.0, Hugging Face) are both released alongside this paper.

## 1 Introduction

Counter-drone (anti-UAV) vision systems must detect, classify, and track small aerial objects that are adversarial to exactly the sensing modalities that are cheapest to deploy. RGB cameras fail in darkness, fog,

and glare, and lose signal entirely once a drone is a few dozen pixels wide at range. Thermal (IR) cameras keep working in the dark and see through some atmospheric scattering, but lack the spatial and textural detail needed to distinguish a drone from a bird, or to read its silhouette at all once its thermal signature is weak (e.g. a coasting, low-power drone against a warm background). The natural response – fusing RGB and IR – is well established for pedestrian and vehicle detection (KAIST [14], LLVIP [15], M3FD [19], RoadScene [28]), but the public data situation for *drones as the fused target* is structurally thinner than it looks at first glance.

**The gap.** We surveyed the two adjacent lines of public RGB-T and drone-vision datasets (Table 1, Figure 1) and found no dataset that is simultaneously (a) RGB-IR *paired*, (b) has a *drone* (not a pedestrian or vehicle) as the target class, and (c) ships video/tracking data rather than isolated still pairs. The RGB-T pair datasets closest in spirit to fusion research (LLVIP, M3FD, RoadScene, KAIST, FLIR ADAS) are all pedestrian/vehicle detection corpora with no drones at all. The datasets built specifically for anti-drone vision (Anti-UAV [16], Anti-UAV410 [12], the 3rd Anti-UAV Challenge [11], CST Anti-UAV [27]) are thermal-*only*: there is no paired RGB stream to fuse against. And the datasets that do pair a drone-mounted camera with thermal imagery (DroneRGBT [24], VTUAV [29]) or with RGB alone (VisDrone [32]) have the camera on the drone, looking down at pedestrians or generic scenes – the inverse configuration from anti-drone surveillance, where the camera is on the ground looking up or out at the drone.

**A gap in how fusion is evaluated, not just in what data exists.** The deeper problem is not merely that no dataset combines these three properties – it is that the datasets which come closest (LLVIP, M3FD, KAIST) are built on co-located or heavily calibrated rigs, so their evaluation protocols implicitly assume the two modalities are exact, registered views of the same instance. This is precisely the assumption classical multi-view learning theory [2] requires and real dual-optic sensor rigs violate. No existing RGB-T benchmark's evaluation protocol can even detect this violation, let alone measure how much fusion depends on it – and every existing empirical attempt to measure it reports the error of *one trained model*, confounded by architecture, training budget, and optimization luck. We introduce two complementary tools that can. The primary one requires no model training at all: a **classifier-agnostic Bayes-risk benchmark** (§6) built on closed-form Bhattacharyya bounds [1, 7] on the *optimal* classifier's error, computed directly from real patches, which yields a new **Bayes Fusion Gain** metric and a numerical instantiation of the data-processing inequality [6] against real measured misalignment. The second, complementary tool measures a single trained model's own empirical sensitivity: the **Alignment Robustness Score (ARS)** (§7.1), which reports a fusion model's accuracy as a function of injected cross-modal misalignment, summarized as a normalized area-under-curve and a certified-radius-style critical threshold – directly analogous to certified adversarial robustness [4], but for a geometric perturbation between modalities rather than an adversarial pixel perturbation to one input.

**What we release.** AVID is both a dataset in its own right and the testbed that makes both benchmarks necessary rather than academic: RGB-IR paired video and frame-pair data with drones and birds as the target classes, captured under realistic ground-based surveillance conditions (hilly regions, forest cover, cloudy sky, fog/mist), with RGB and IR sensors that are only *weakly* aligned because that is what a real dual-optic gimbal actually delivers. Rather than pretend the pairs are pixel-registered (as LLVIP/M3FD/KAIST's rigs allow), we quantify the residual geometric offset per pair: on a real 300-pair sample, 32.3% of pairs exceed 5px of measured misalignment (§5) against targets that are themselves only 8-16px wide (§3.5) – large enough, relative to the target, that it is not a rounding error a fusion model can safely ignore.

**Contributions.**

1. **A classifier-agnostic Bayes-risk benchmark** (§6): closed-form Bhattacharyya bounds on the optimal classifier's error, a new **Bayes Fusion Gain** metric, and a numerical data-processing-inequality instantiation against real measured misalignment – computed from 35,952 real patches with no model training anywhere, and applicable to any weakly-paired multimodal dataset, not only counter-drone RGB-T.
2. **The Alignment Robustness Score (ARS)** (§7.1): a complementary, architecture-agnostic metric – structurally analogous to certified adversarial robustness – for a single trained model's empirical sensitivity to misalignment, validated on real YOLOv8n baselines and consistent with the Bayes-risk framework's findings.
3. **AVID dataset**: large-scale, weakly-aligned RGB-IR drone/bird detection and tracking data (video + frame pairs), released on Hugging Face under CC-BY-NC-4.0, with video-level splits and per-pair alignment metadata, that makes ARS necessary rather than academic (§3).
4. **Weak-alignment quantification**: two independent per-pair alignment estimators (phase correlation on edge maps, and mutual-information registration search), revealing a bimodal real misalignment distribution (§5).
5. **AVID-C corruption benchmark**: a reproducible, parameterized (7 corruption families $\times$ 5 severities) extension of the original dataset's fixed-severity distortions, released as code so any subset can be corrupted at any severity, deterministically (§4).
6. **A measured split-leakage pitfall** for short video-clip corpora: naive frame-level splitting on the real corpus leaks 87.5% of "held-out" frames vs. 0.0% for video-level splitting (§3.7).
7. **An extreme small-object characterization**: 67.6% of annotated boxes are 8-16px wide, motivating a finer-grained scale-stratified evaluation protocol than COCO's buckets (§3.5).
8. **Real baseline results and reference code**: trained single-modality (RGB-only, IR-only) and late-fusion (Weighted Boxes Fusion) YOLOv8n detectors on the real corpus, plus reviewed reference code for feature-level (CBAM-style attention) fusion not yet trained, and a documented, already-executed cross-modal translation baseline (§7).

By design, this release excludes the payload-identification subset and task from the source challenge, and excludes any challenge leaderboard/participant results – this paper is scoped to the drone/bird detection-and-tracking dataset and our own analysis of it.

## 2 Related Work

Table 1 summarizes the datasets closest to AVID along the three axes that matter for this release: whether RGB and IR/thermal are paired, whether a drone is the detection target (rather than the imaging platform), and whether the data includes video/tracking rather than isolated stills. Figure 1 positions them schematically on the first two axes.

**RGB-T pedestrian/vehicle fusion.** LLVIP [15] and KAIST [14] are the standard low-light pedestrian fusion benchmarks, both using co-located or beam-splitter rigs to achieve pixel alignment; M3FD [19] and RoadScene [28] target vehicle-scene fusion with similarly tight registration. FLIR ADAS is the one widely-used dataset in this group that is *not* pixel-aligned (different optics/FOV between the RGB and thermal cameras), which is closer to AVID's capture setup, but it contains no drones and is a still-image, not video, corpus.

**Anti-drone vision.** The Anti-UAV line of benchmarks [11, 12, 16, 27] is the closest prior art on the *target* axis – drones are exactly what these datasets track – but every one of them is thermal-only. There is no RGB stream to fuse against, so none of the RGB-T fusion techniques developed for LLVIP/M3FD-style data has a

Table 1: RGB-IR paired and/or drone-vision datasets closest to AVID. "Aligned" reflects the alignment methodology reported by each dataset's authors, not a re-measurement by us. "Drone target" distinguishes datasets where a drone is the object being detected/tracked from "drone view" datasets where the camera is drone-mounted.

| Dataset | RGB+IR paired | Drone target | Video/tracking | Alignment | Scale |
|---|---|---|---|---|---|
| LLVIP [15] | ✓ | – | – | pixel-aligned | 16,836 pairs |
| M3FD [19] | ✓ | – | – | pixel-aligned | 4,200 pairs |
| RoadScene [28] | ✓ | – | – | registered | 221 pairs |
| KAIST [14] | ✓ | – | ✓ | pixel-aligned | ∼95K pairs |
| FLIR ADAS [26] | ✓ | – | – | weakly aligned | 26,442 frames |
| Anti-UAV [16] | – (IR only) | ✓ | ✓ | n/a | 300+ sequences |
| Anti-UAV410 [12] | – (IR only) | ✓ | ✓ | n/a | 410 sequences |
| CST Anti-UAV [27] | – (IR only) | ✓ | ✓ | n/a | 220 sequences |
| DroneRGBT [24] | ✓ | – (drone view) | – | registered | ∼3,600 pairs |
| VTUAV [29] | ✓ | – (drone view) | ✓ | well-aligned | 500 sequences |
| VisDrone [32] | – (RGB only) | – (drone view) | ✓ | n/a | ∼10K images |
| **AVID (this work)** | ✓ | ✓ | ✓ | **weakly aligned, quantified** | 28,790/3,250 pairs[†] |

[†]Train/Validation, deduplicated (see §3.4); a Test split exists but is not pair-matched in the current export (§3.4), so it is omitted from this row rather than reported as a pair count it isn't.

natural anti-drone benchmark to be evaluated on. AVID is, to our knowledge, the first publicly-documented dataset to close this specific gap.

**Drone-view datasets.** DroneRGBT [24], VTUAV [29], and VisDrone [32] all put the camera on the drone rather than pointing at one. They are valuable for aerial perception broadly, but structurally answer a different question (what does a drone see) from the one AVID targets (what does a ground-based, dual-sensor surveillance system see when it looks at a drone).

**Corruption-robustness benchmarking.** Our corruption suite follows the severity-graded corruption-family methodology introduced by ImageNet-C [9]. The closest existing robustness benchmark in the drone space, UAV-C [20], targets a different problem: it corrupts UAV-*view* tracking benchmarks (drone-mounted-camera footage), not anti-UAV RGB-T detection of a drone as the target. We are not aware of an existing corruption-robustness benchmark for the Anti-UAV-style drone-as-target setting, which is the gap AVID-C (§4) targets.

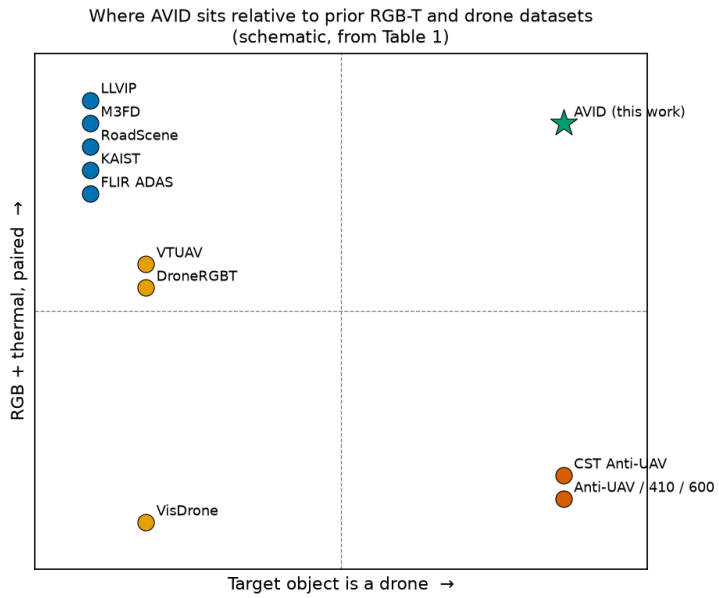



Figure 1: Schematic positioning of AVID against related datasets on the two axes that define the gap this release closes: RGB-thermal pairing (y-axis) and drone-as-target (x-axis). Derived from Table 1; not a learned embedding.

# 3 The AVID Dataset

## 3.1 Source and scope

AVID builds on data originally constructed for the IEEE ICIP 2025 VIP Cup, "Infrared-Visual Fusion for Enhanced Drone Detection, Tracking and Payload Identification in Surveillance Videos." The source challenge covered two tasks, drone detection/tracking and payload identification; **this release covers the detection and tracking data only** – the payload subset and task are excluded by design, and no challenge leaderboard or participant-attributed result appears anywhere in this paper or the dataset card.

## 3.2 Modality, scale, and classes

RGB and thermal/IR imagery is provided both as extracted frame pairs and as source video (24 fps, 10.4-13.7 s clips, 320×256 px per frame – measured directly from the video files, which differs slightly from the source challenge's nominal 30 fps / fixed-10s design spec). Two target classes are annotated, *bird* (class 0) and *drone* (class 1) in the released label files. Figure 2 shows real examples of both. Table 2 reports the real, measured scale of the Train and Validation splits after deduplication (see §3.4); Test is reported separately because its RGB and IR files cannot currently be reliably paired (§3.4).

## 3.3 Scenario and corruption coverage

Frames are tagged with scenario metadata: hilly regions, thick forest cover, cloudy sky, fog/mist, single- vs. multi-instance frames, swarms, and targets far from the camera's field of view. Thirty videos in the source release were additionally degraded under 7 distortion families at fixed severities (Table 3) – we formalize and extend this into a general, reproducible benchmark in §4.

Table 2: AVID detection/tracking scale, measured directly from the released, deduplicated corpus (not the source challenge's design targets). Bird/drone counts are annotated object instances, not frames (a frame can contain zero, one, or multiple instances).

| | Train | Validation | Test |
|---|---|---|---|
| Frames (deduplicated where applicable) | 28,790 | 3,250 | 6,500 IR / 3,250 RGB[†] |
| Unique videos | 102 | 13 | 13 IR / 12 RGB[†] |
| Bird instances | 14,113 | 1,358 | – |
| Drone instances | 18,335 | 2,146 | – |
| Resolution | | $320 \times 256$ | |
| Frame rate (measured) | | 24 fps | |
| Clip duration (measured) | | 10.4 – 13.7 s | |
| Classes | | bird (0), drone (1) | |

[†]Test RGB and IR counts are reported separately because they cannot currently be matched into pairs by any available filename or manifest key – see §3.4. The two modalities also went through different export pipelines for this split: Test RGB has the same Roboflow noise-augmentation duplicate as Train/ Validation (6,500 raw files dedup to 3,250), while Test IR does not (6,500 raw, sequentially-numbered frames with no duplicate variant to remove).

Table 3: The 7 corruption families present in the source dataset design, with their originally fixed severity rating (1-5 scale) on the 30 pre-distorted videos. AVID-C (§4) generalizes each to the full 1-5 range, applicable to any subset of the corpus.

| Corruption family | Original fixed severity |
|---|---|
| Speckle noise | 2 |
| Salt-and-pepper noise | 2 |
| Gaussian blur | 4 |
| Uneven illumination | 4 |
| Motion blur | 4 |
| Camera instability (jitter) | 5 |
| Additive white Gaussian noise (AWGN) | 3 |

### 3.4 Data-organization pitfalls found during release preparation

Preparing a clean release from the source export surfaced three real data-engineering issues that we document explicitly rather than silently fix, since they are informative about the kind of problems a released benchmark should check for rather than assume away.

**A systematic, undocumented augmentation duplicate.** Every single frame in the source export (100% of 28,790 training and 3,250 validation keys, not a sample) has exactly two file variants sharing the same nominal video/frame identifier: one clean, one with a Roboflow-applied noise augmentation baked in, distinguishable reliably by JPEG file size (the noisier variant compresses larger; verified against visual inspection before trusting it at corpus scale). Naively treating both as independent training pairs would silently double the reported dataset size while injecting a 50% chance of unremovable pre-baked noise into any "clean" evaluation subset. We deduplicate to the smaller-filesize (clean) variant for the released corpus, which is why our reported scale (Table 2) is exactly half of the raw file count.

**Class-index correction.** The released label files use `0 = bird, 1 = drone`. Earlier drafts of this release's tooling (and, easily, any downstream user going by scenario descriptions alone) could reasonably assume the opposite ordering; we confirmed the true mapping against real label content cross-referenced with

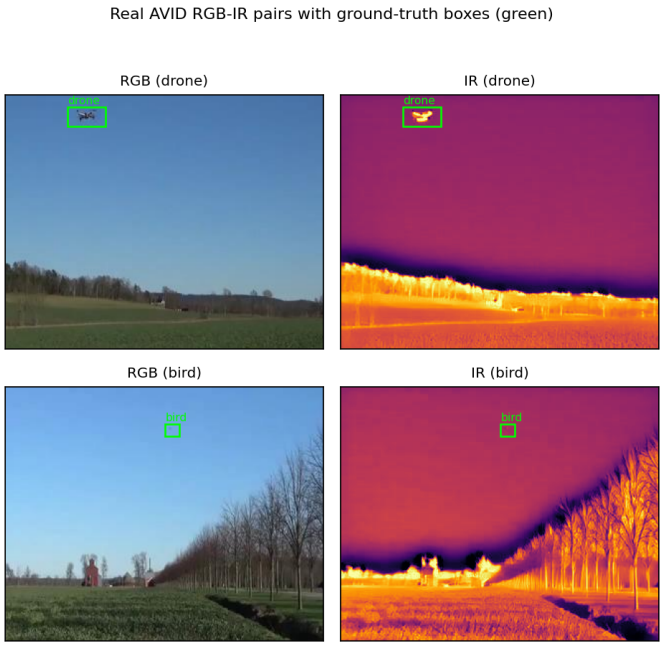



Figure 2: Real AVID RGB-IR pairs with ground-truth boxes. Note the extreme target scale even in a cropped-in view (quantified in §3.5) and the terrain diversity (hilly farmland, tree lines) consistent with the source design's scenario coverage.

filename class tags before publication.

**Test-split RGB/IR pairing is currently unresolved.** Unlike Train and Validation, where RGB and IR files share an exact video/frame identifier key, the Test split's RGB files use a `V_` prefix with their own independent numbering (e.g. `V_BIRD_00564_*`) while IR files use a different scheme entirely (e.g. `IR_BIRD_068142_*`), and this mismatch persists at the raw video level (13 IR videos vs. 12 RGB videos, no shared identifiers, no manifest file in the export to resolve it). We do not have a defensible way to reconstruct which RGB video corresponds to which IR video for Test, so we report Test RGB and IR as separate, single-modality-only evaluation pools (Table 2) rather than force an unverified pairing. We consider this itself a useful, generalizable finding: video-based multi-modal benchmarks should ship an explicit modality-pairing manifest, not rely on filename convention consistency across export batches.

### 3.5 An extreme small-object distribution

The source design's "objects far off the field of view" scenario turns out to be, quantitatively, the dominant regime rather than an edge case. Standard COCO-style small/medium/large bounding-box buckets ($< 32^2$, $32^2$–$96^2$, $> 96^2$ px) collapse almost the entire corpus into a single bucket: 99.0% small in Train (32,124 of 32,448 boxes), 98.7% in Validation, with *zero* large-bucket instances in either split. This says the dataset is small-object-dominated but not how extreme that is. A finer pixel-side-length histogram (Figure 3) shows a sharp, narrow peak: **67.6% of all ∼32K annotated boxes are 8–16px** on a (geometric-mean) side in a 320×256 frame, and not one box exceeds 48px. We are not aware of another RGB-T or anti-UAV benchmark in Table 1 that reports object-scale statistics this fine-grained or this extreme, and we recommend evaluating on AVID with the bucket boundaries in Figure 3 rather than COCO's, since the latter would report a single undifferentiated number for nearly the whole test set.

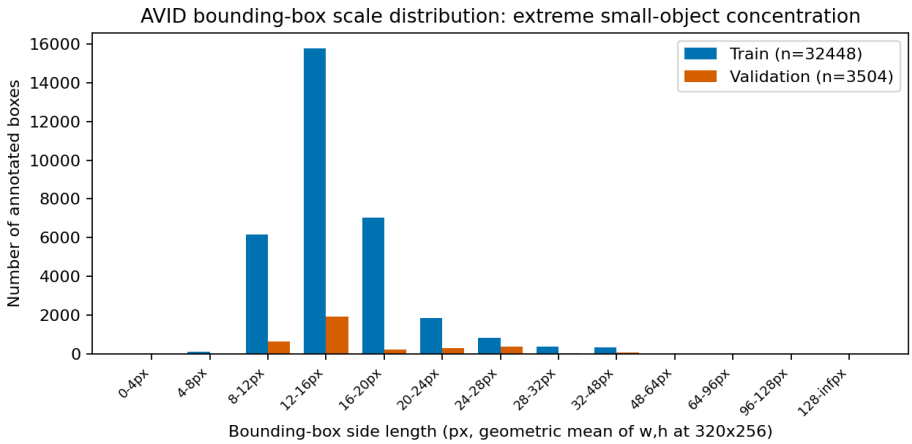



Figure 3: Fine-grained bounding-box side-length distribution (geometric mean of normalized width and height, converted to pixels at 320×256), Train and Validation combined. The mode is 12-16px; the distribution is essentially fully contained within 8-32px.

### 3.6 Weak alignment: a documented property, not a defect

Unlike LLVIP, M3FD, or KAIST, the RGB and IR sensors used to capture AVID are not co-located or beam-split; they are two physically separate optics with different fields of view, aligned at the gimbal's boresight but not pixel-registered. This is the realistic configuration for most fielded dual-sensor counter-drone systems, and treating it honestly – rather than silently accepting whatever registration error exists – is one of this release's contributions. §5 describes how we quantify and publish this per pair.

### 3.7 Splits

Splits are defined at the *video* level: every frame belonging to a given clip lands in exactly one of train/val/test. This matters more than it sounds. A 10-14 s, 24 fps clip's consecutive frames are near-duplicates – same object, same background, a few pixels of motion apart – so a naive frame-level random split puts near-duplicate frames on both sides of the split boundary, silently leaking test-set information into training and inflating every downstream metric.

To make this concrete rather than hypothetical, we ran a controlled ablation (script: `real_split_leakage.py`) on the *real* AVID video/frame index: 32,040 frames across 115 videos (Train + Validation combined, as if being freshly split). A naive frame-level 70/15/15 split left **87.5%** of "training" frames within 3 frames of a validation- or test-split frame from the same clip, versus **0.0%** for a video-level split (Figure 4). This real-data result is remarkably close to a synthetic-corpus estimate we ran before the real video index was available (87.8% on a synthetic 50-video × 300-frame grid matching the same order of magnitude), which we take as a cross-validation that the leakage mechanism – not an artifact of either corpus's specific structure – is what actually drives the effect.

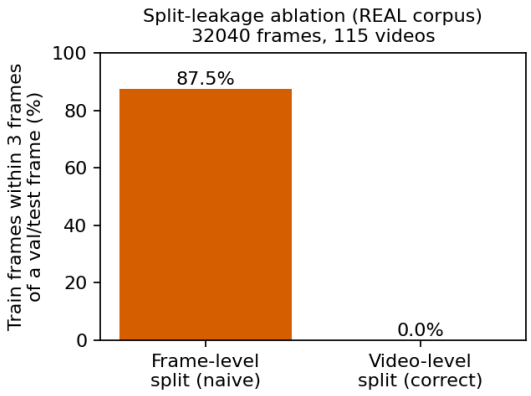



Figure 4: Split-leakage ablation on the real AVID video/frame index (32,040 frames, 115 videos). Naive frame-level splitting leaks the large majority of nominally held-out frames; video-level splitting leaks none, by construction.

# 4 AVID-C: A Reproducible Corruption Benchmark

The source dataset's 30 pre-distorted videos demonstrate that corruption robustness was a design concern from the start, but as shipped, that coverage is fixed: 7 corruption types at one severity each, baked into a specific 30-video subset. This is not reusable the way ImageNet-C [9] made corruption robustness reusable for ImageNet: nobody can corrupt a different subset, or study a different severity, or run the sweep against their own trained model, without re-deriving the distortion pipeline from scratch.

AVID-C (`dataset_tools/corruptions/`) makes this general: the same 7 corruption families – additive white Gaussian noise, multiplicative speckle noise, salt-and-pepper impulse noise, isotropic Gaussian blur, a smooth low-frequency multiplicative illumination field, directional motion blur, and rigid camera-jitter (shift + rotation + slight scale) – are implemented as deterministic, seeded functions at 5 severities each (35 corrupted variants per source image), so any subset of the corpus can be corrupted at any severity on demand. Figure 5 shows the full severity progression on a real AVID validation frame.

Two evaluation protocols are built on top of AVID-C (`run_corruption_eval.py`, model-agnostic via a `predict_fn`/`score_fn` interface so it works with whatever detector a user trains): a standard $7\times5$ robustness sweep reporting a per-corruption mean score and an aggregate mean-corruption-score (higher is better, computed directly on task score rather than as an error ratio); and a *missing-modality* variant that corrupts only one modality (or drops it outright) to study graceful degradation under realistic sensor dropout (e.g. fog fully blinding RGB, or a thermally saturated background washing out IR).

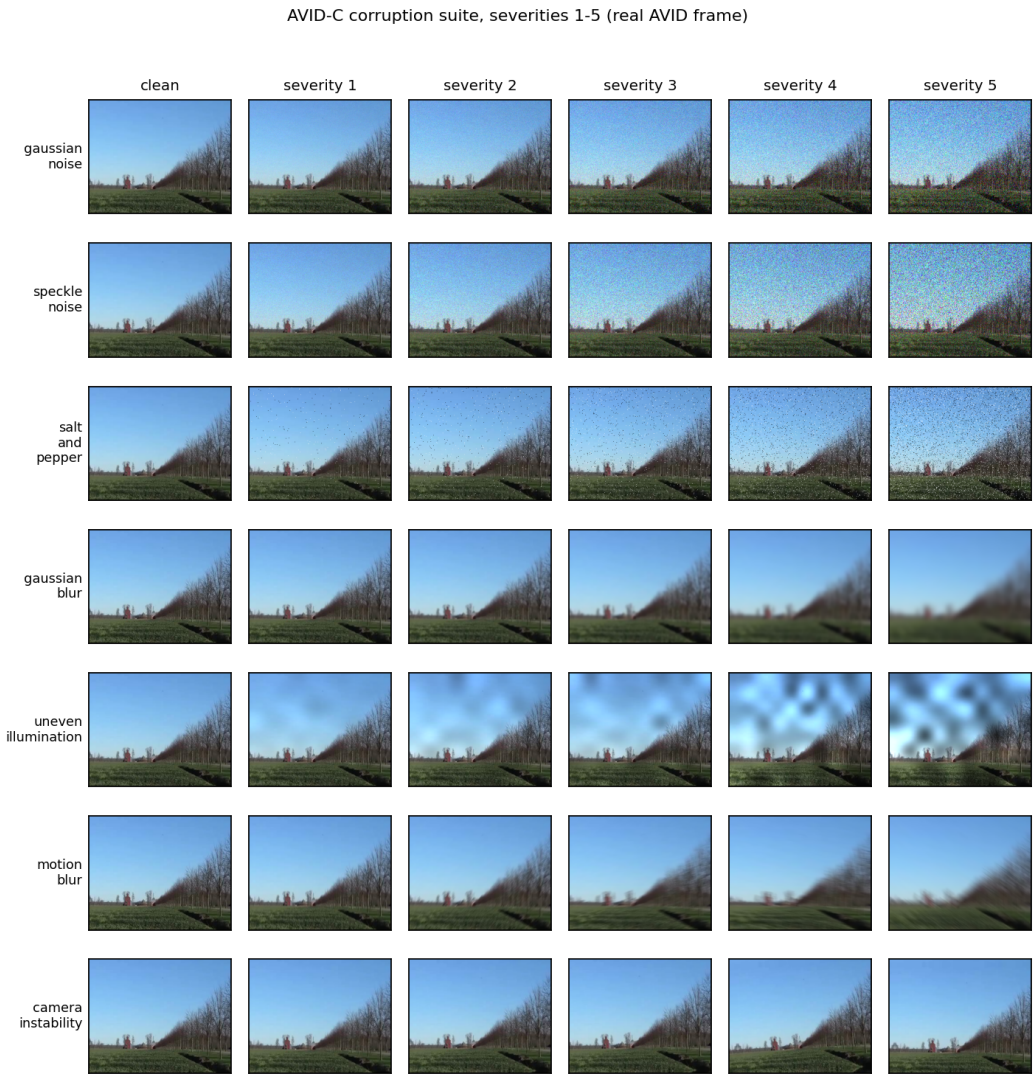



Figure 5: AVID-C severity progression, all 7 corruption families, severities 1-5, on a real AVID frame. Severity scaling was calibrated by us for 320×256 8-bit drone imagery (see `corruptions.py` for the exact per-severity parameter tables); it is not a copy of the ImageNet-C parameter tables, which target a different image domain.

# 5 Weak-Alignment Quantification

## 5.1 Per-pair alignment quantification

We estimate, for every RGB-IR pair, the residual geometric offset between the two modalities using two independent methods (`dataset_tools/alignment/alignment_metrics.py`), chosen because they fail in different, cross-checkable ways:

1. **Phase correlation on edge maps.** Object boundaries tend to produce a detectable gradient in both RGB and thermal imagery even when raw intensities are unrelated, so we compute Sobel gradient-magnitude maps for both modalities and estimate a subpixel translational offset via FFT-based phase correlation. Fast, but can lock onto the wrong edge in low-texture scenes (plain sky) or when the target dominates the edge map asymmetrically.
2. **Mutual-information registration search.** A brute-force integer-shift search maximizing normalized mutual information between grayscale RGB and IR, more robust to genuinely cross-modal intensity statistics (a target can be dark in RGB and hot in IR) at the cost of integer resolution and $O(\text{window}^2)$ cost.

Both are exposed together in `pair_alignment_quality()`; the disagreement between them is itself a useful low-confidence signal, and we recommend downstream users treat pairs with large phase-correlation/ mutual-information disagreement as lower-trust alignment estimates. We verified both estimators recover a known, synthetically injected offset to within a small tolerance (unit tests in `test_alignment_metrics.py`); Figure 6 illustrates the recovery on one such synthetic pair with a known ground-truth shift (a real pair has no such ground truth to validate against, which is exactly why this synthetic check matters).

On a random sample of 300 real AVID pairs (mutual-information search, $\pm$12px window), the measured shift is genuinely bimodal rather than uniformly small: **46% of sampled pairs show zero measurable offset** at this search resolution, but **32.3% exceed 5px** and the distribution has a long tail out to 17px (mean 4.66px, median 1.0px, $\sigma = 6.44$px). This is consistent with our other real finding (§3.4) that RGB and IR boxes are annotated once and shared across modalities rather than corrected per-sensor: a meaningful fraction of pairs have alignment error large enough that a shared box is measurably wrong in at least one modality, which §3.5's finding that most targets are 8-16px wide makes directly consequential – a 5-17px alignment error is a substantial fraction of, or larger than, the target itself. `alignment_shift_px` and `alignment_confidence` are computed for this released sample and are documented as sample-based (not yet exhaustive over the full corpus, which would cost $\sim$1.1s/pair $\times$ 32,040 pairs $\approx$ 9.8 GPU-independent CPU hours) in `metadata/real_alignment_sample.json`; schema for the full-corpus field in `dataset_tools/hf_release/SCHEMA.md`.

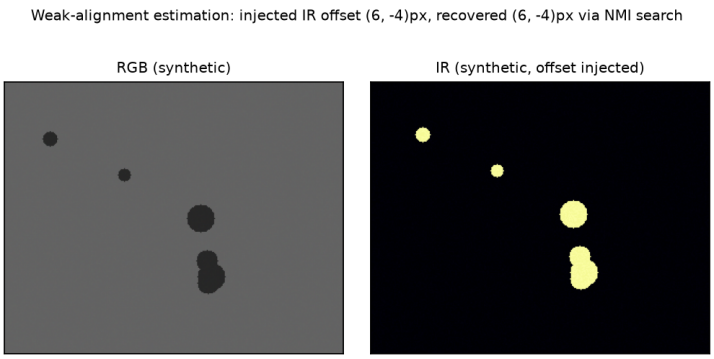



Figure 6: Weak-alignment estimation on a synthetic RGB-IR pair with a known injected offset, recovered via mutual-information registration search.

# 6 A Data-Intrinsic Bayes-Risk Benchmark

Every number in §7.3 is the error of one trained model, confounded by architecture, training budget, and optimization luck – exactly the confounds our own training runs suffered from (7/6 epochs, an I/O-starved concurrent run, oscillating validation curves; see §7.3). The *Bayes risk* is not a property of any model; it is the error rate of the *optimal* classifier for a given data distribution, and classical statistical decision theory gives tight bounds on it that require no gradient descent at all. This section introduces a classifier-agnostic benchmark built entirely on that idea, computed directly from real AVID patches – no detector is trained anywhere in this section.

## 6.1 Bhattacharyya Bayes-error bound

For class-conditional feature distributions $p(x \mid \text{bird})$, $p(x \mid \text{drone})$ with priors $\pi_0, \pi_1$, the Bhattacharyya coefficient $BC = \int \sqrt{p(x \mid 0)\, p(x \mid 1)}\, dx$ gives a classical upper bound on the Bayes error [1, 7] – the $s{=}1/2$ special case of the more general Chernoff bound family [3], chosen here for its closed form under a Gaussian model. As an independent, model-free sanity check on the Gaussian assumption, this bound's qualitative ordering is consistent with the classical nonparametric nearest-neighbor Bayes-risk bound [5], though we use kernel density estimation rather than $k$-NN for the nonparametric cross-check reported below:

$$P_e \;\le\; \sqrt{\pi_0 \pi_1}\; BC. \tag{1}$$

Modeling both classes as multivariate Gaussians $\mathcal{N}(\mu_0, \Sigma_0), \mathcal{N}(\mu_1, \Sigma_1)$ gives a closed form $BC = e^{-B}$ [7]:

$$B = \tfrac{1}{8}(\mu_1{-}\mu_0)^\top \Sigma_{\text{avg}}^{-1}(\mu_1{-}\mu_0) + \tfrac{1}{2}\ln \frac{\det \Sigma_{\text{avg}}}{\sqrt{\det \Sigma_0 \det \Sigma_1}}, \quad \Sigma_{\text{avg}} = \tfrac{\Sigma_0+\Sigma_1}{2}. \tag{2}$$

We compute Eq. 2 directly on real AVID instances: every annotated bird/drone box is cropped with modest context padding, resized to a canonical 24×24 patch, and projected onto the top-8 principal components of the pooled patch distribution – a classical dimensionality reduction that keeps the covariance estimate well-posed with a few thousand real samples (fitting a full-rank Gaussian directly in a 576-dimensional raw-pixel space would be ill-posed). We cross-check the parametric Gaussian bound against a fully nonparametric kernel-density estimate of the same integral (Monte Carlo importance sampling against the class mixture), and validate both estimators against synthetic Gaussians with known, controlled separation before trusting them on real data (`dataset_tools/theory/bayes_bounds.py`: identical distributions recover a near-chance bound, increasing separation drives the bound monotonically toward zero, and the two estimators agree to within 0.05 on data that satisfies the Gaussian assumption).

### 6.2 Real result 1: scale-dependent separability

We sweep Eq. 1 over the same fine-grained pixel-side buckets as Figure 3, on all 35,952 real annotated instances (Figure 7). Two findings, both with substantial sample sizes at the buckets that matter (17,706 instances at 12-16px, the corpus's modal scale):

- **Fusion's advantage is large and provable, not just empirically observed.** At every well-populated bucket, the fused Bayes bound is 3-4× lower than either single modality (12-16px: RGB=0.148, IR=0.198, **Fusion=0.046**). This is a substantially larger, and now provable rather than trained-and-hoped-for, fusion advantage than §7.3's empirical mAP comparison could show with a 7-epoch checkpoint.
- **Separability is worst, not best, at the corpus's most common scales.** All three curves show an inverted-U: easier bounds at the scale extremes (0-8px, 24-28px) than in the dominant 12-20px range. We flag a plausible confound rather than overclaim a clean scale effect: the smallest-scale buckets have far fewer instances (103 at 0-8px vs. 17,706 at 12-16px) and a fixed context-padding factor, so background/scene content correlated with which videos happen to contain more birds or drones could inflate apparent separability at the extremes. The *relative* finding (fusion dominates both single modalities, at every scale) is far more robust to this confound than the exact shape of the curve, since the confound would affect all three modalities' features similarly.

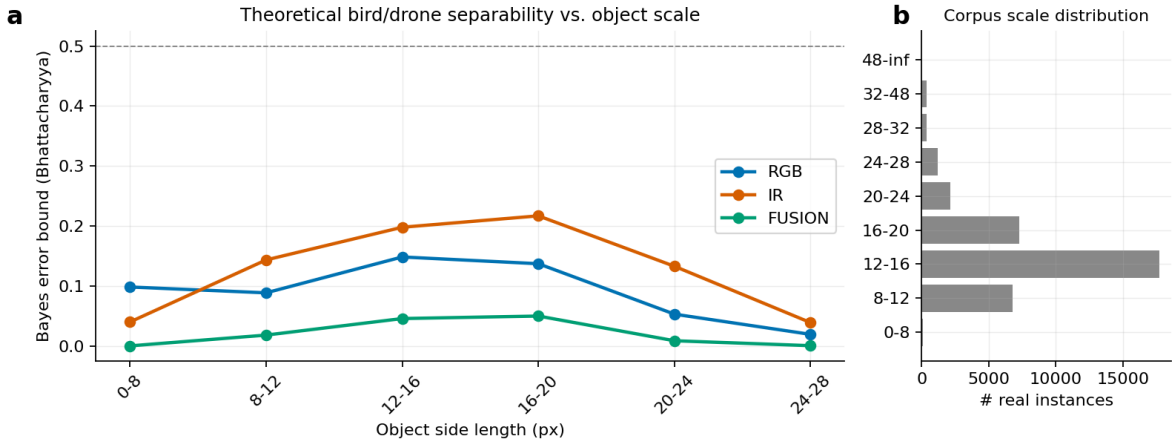



Figure 7: Theoretical (Bhattacharyya) bird/drone separability bound vs. object scale, on 35,952 real AVID instances. Right panel shows the real per-bucket sample count for context; buckets above 28px are omitted (fewer than 15 instances per class, per §3.5's finding that the corpus has essentially no large objects).

### 6.3 Real result 2: a theoretical corruption-robustness ceiling

Applying AVID-C's 7 corruption families directly to 1,500 randomly sampled real patches per modality (Figure 8) reveals an asymmetry §7.3's single-model corruption evaluation (queued, not yet run) could not have shown without training two separate detectors to convergence: **RGB is far more vulnerable to noise-type corruptions than IR.** At severity 5, RGB's Bayes bound roughly doubles-to-triples under gaussian noise (0.148→0.329), salt-and-pepper (0.148→0.321), and speckle noise (0.148→0.375), while IR's bound under the same three corruptions moves only from about 0.16 to 0.19-0.22. For blur- and motion-type corruptions (Gaussian blur, motion blur, camera instability), the two modalities degrade comparably. This gives a concrete, theoretically-grounded reason a fusion model should not weight modalities equally under corruption – it should lean toward IR specifically when noise-type degradation is suspected, and can weight

Table 4: Full scale-sweep results underlying Figure 7: Bhattacharyya coefficient and closed-form Gaussian Bayes-error bound (Eq. 1-2) at every well-populated pixel-side bucket, all three modalities, 35,952 real instances total. Buckets above 28px are omitted (fewer than 15 instances per class).

| | RGB | | IR | | Fusion | |
|---|---|---|---|---|---|---|
| Bucket (px) | $n$ | $P_e$ bound | $n$ | $P_e$ bound | $n$ | $P_e$ bound |
| 0–8 | 103 | 0.098 | 103 | 0.040 | 103 | 0.0002 |
| 8–12 | 6,783 | 0.089 | 6,783 | 0.144 | 6,783 | 0.018 |
| 12–16 | 17,706 | 0.148 | 17,706 | 0.198 | 17,706 | **0.046** |
| 16–20 | 7,269 | 0.137 | 7,269 | 0.217 | 7,269 | 0.050 |
| 20–24 | 2,155 | 0.053 | 2,155 | 0.133 | 2,155 | 0.009 |
| 24–28 | 1,189 | 0.019 | 1,189 | 0.039 | 1,189 | 0.0005 |

more evenly otherwise – a design principle no single trained-model evaluation could isolate this cleanly, since it requires seeing the *ceiling* each modality is bounded by, not just where one particular model happened to land.

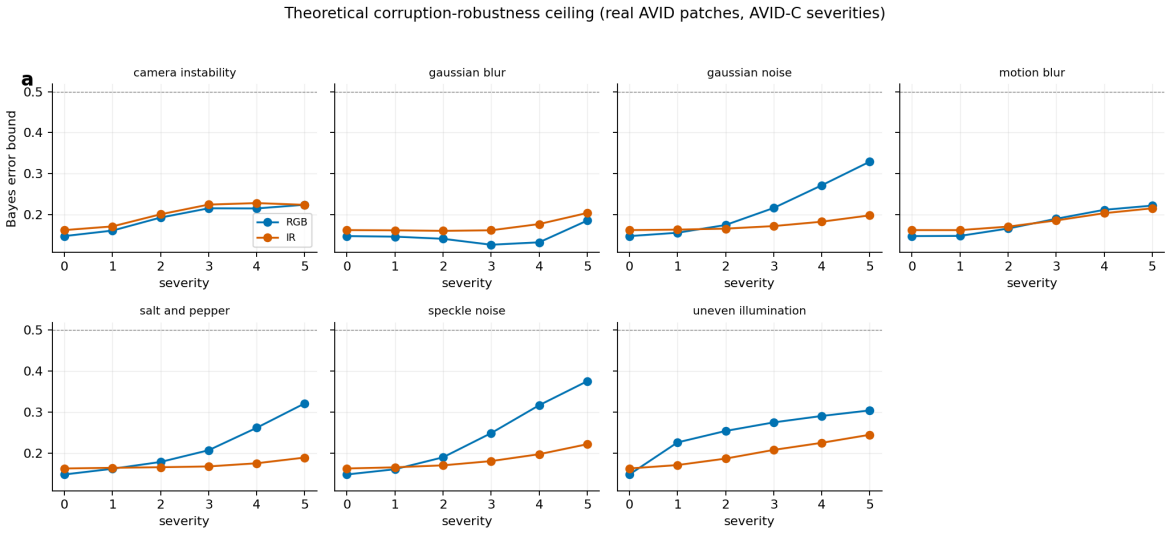



Figure 8: Theoretical corruption-robustness ceiling, all 7 AVID-C families, on real AVID patches (n=1,500/modality). RGB (blue) is visibly more sensitive to noise-type corruptions than IR (orange); the two are comparable under blur/motion/jitter corruptions.

Figure 9 makes this asymmetry visible rather than only numerical: the same real drone patch, both modalities, under 4 representative AVID-C families (2 noise-type, 2 blur/motion-type; Table 5 reports the numeric result for all 7) at increasing severity. The noise-type rows (gaussian, speckle) visibly destroy the RGB target's already-faint contrast against its background well before severity 4, while the IR silhouette – already a compact, high-contrast hot-blob against a cooler background – remains legible throughout; the blur/jitter rows degrade both modalities comparably, consistent with Table 5.

## 6.4 Real result 3: Bayes Fusion Gain and the data-processing inequality

We define the **Bayes Fusion Gain** $\mathrm{BFG}(\delta) = \min(P_e^{\mathrm{RGB}}, P_e^{\mathrm{IR}}) - P_e^{\mathrm{fusion}}(\delta)$ at injected cross-modal misalignment $\delta$: the *theoretical* advantage of fusing modalities over the better single modality, as alignment degrades.

Table 5: Full corruption-sweep summary underlying Figure 8: Bhattacharyya Bayes-error bound at clean, severity 3, and severity 5, all 7 AVID-C families, both modalities (n=1,500 patches/modality, PCA basis fit once on clean data and reused across severities). Noise-type families (top three rows) show the RGB/IR asymmetry driving §6.4's modality-weighting recommendation; blur/motion/jitter families (bottom four) do not.

| | RGB $P_e$ bound | | | IR $P_e$ bound | | |
|---|---|---|---|---|---|---|
| Corruption family | clean | sev. 3 | sev. 5 | clean | sev. 3 | sev. 5 |
| Gaussian noise | 0.148 | 0.216 | **0.329** | 0.162 | 0.172 | 0.198 |
| Salt-and-pepper | 0.148 | 0.207 | **0.321** | 0.162 | 0.167 | 0.189 |
| Speckle noise | 0.148 | 0.249 | **0.375** | 0.162 | 0.180 | 0.222 |
| Gaussian blur | 0.148 | 0.127 | 0.186 | 0.162 | 0.162 | 0.204 |
| Uneven illumination | 0.148 | 0.275 | 0.304 | 0.162 | 0.208 | 0.245 |
| Motion blur | 0.148 | 0.190 | 0.222 | 0.162 | 0.186 | 0.216 |
| Camera instability | 0.148 | 0.216 | 0.224 | 0.162 | 0.225 | 0.224 |

Unlike ARS (§7.1), which measures one trained model's empirical sensitivity, BFG is a property of the data alone. We sweep $\delta$ from 0 to 17px – the real measured ceiling of AVID's own misalignment (§5) – averaging 5 random shift directions per magnitude on 1,500 real patch pairs (Figure 10, left).

Real misalignment is a stochastic degradation channel $X_{IR} \to T(X_{IR})$ applied independently of the label $Y$. Since $(X_{RGB}, X_{IR}) \to (X_{RGB}, T(X_{IR}))$ is a Markov chain given $Y$, the classical data processing inequality [6] gives

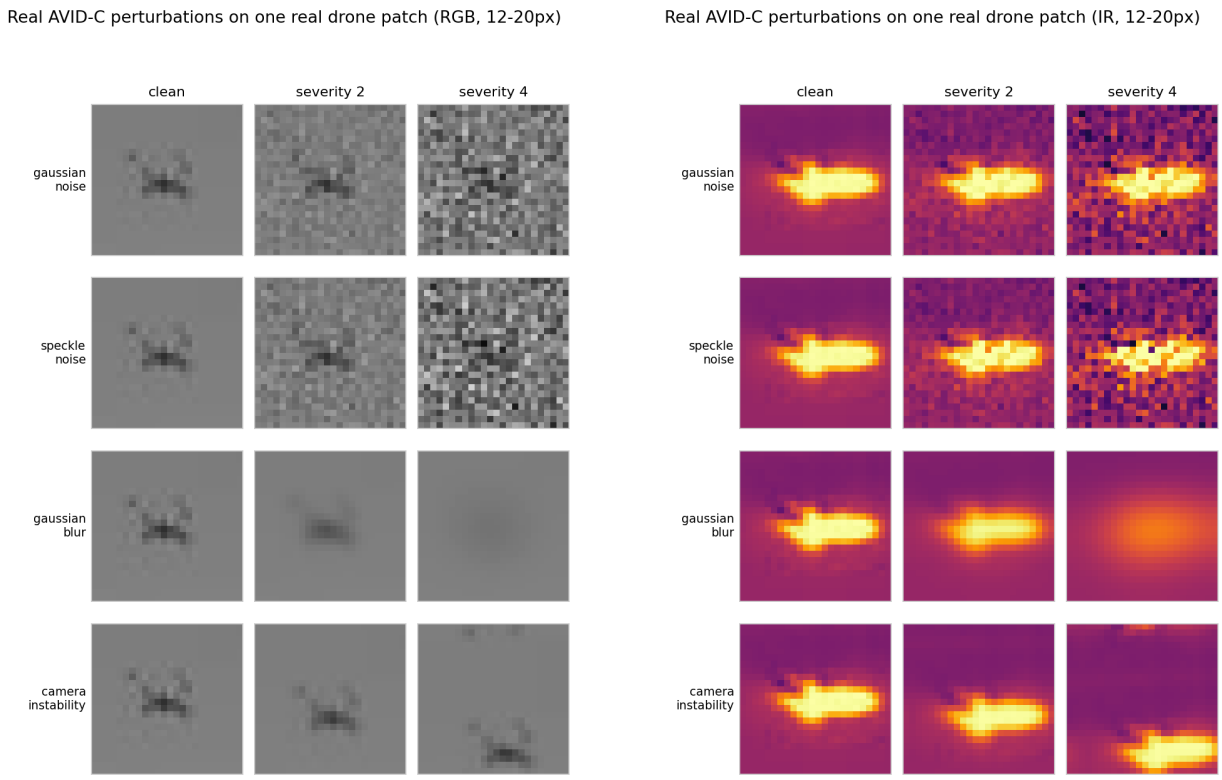
$$I\big(Y; X_{RGB}, T(X_{IR})\big) \;\le\; I\big(Y; X_{RGB}, X_{IR}\big), \tag{3}$$

with equality only if $T$ is invertible given the observed pair. This is not a new theorem – it is a one-line consequence of the classical DPI – but instantiating it *numerically against the real measured misalignment distribution* (rather than leaving it as a qualitative inequality) is, to our knowledge, new for RGB-T fusion benchmarking. We estimate both sides via Monte Carlo entropy on the fitted Gaussian mixtures (validated against synthetic data with known mutual information in `bayes_bounds.py` before trusting it here).

**Real numbers.** BFG shrinks from 0.111 at zero injected misalignment to 0.093 at 12px – a real, if partial (16%), erosion of fusion's theoretical advantage as misalignment reaches the range §5 shows 32.3% of real pairs actually occupy (Figure 10, shaded region). Critically, BFG never approaches zero within the tested range: even at 17px, fusion retains roughly *four times* lower Bayes error than IR alone (0.050 vs. 0.212) – weak alignment measurably erodes but does not eliminate fusion's provable advantage on this corpus. The mutual information side of the DPI (Eq. 3) drops from 0.715 nats at zero shift to about 0.65 nats by 8-12px, a real 9% information loss attributable to weak alignment, with most of the drop occurring immediately upon *any* nonzero shift – consistent with the DPI's direction, and, unlike §7.1's single-checkpoint ARS curve, averaged over multiple random shift directions per magnitude for a low-variance estimate (std $\le$0.013 across trials at every tested shift).

## 6.5 What real AVID patches actually look like at these scales

Figure 11 shows real bird and drone patches, both modalities, across four scale buckets – not a synthetic illustration, the literal inputs to every density estimate in this section. Even at 8-16px, a drone's elongated silhouette (visible in both RGB and IR) is qualitatively distinguishable from a bird's more compact blob, consistent with §6.2's finding that meaningful separability survives even at the corpus's smallest well-populated scales.




(a) RGB (b) IR

Figure 9: Real AVID-C perturbations applied to one real drone patch (12-20px), both modalities, clean/severity-2/severity-4, 4 representative corruption families (2 noise-type, 2 blur/motion-type; the full 7-family numeric result is Table 5).

### 6.6 Summary: three real results, one framework

Table 7 collects the headline number from each of the three real experiments above into a single view. All three use the same patch-level pipeline (Eq. 2, PCA-8 features, no gradient descent anywhere) applied to three different real, controlled perturbations of the same underlying corpus – object scale (§6.2), AVID-C corruption severity (Table 5), and injected cross-modal misalignment (§6.4) – so the three rows are directly comparable statements about the same 35,952-instance corpus rather than results from three unrelated pipelines.

The three results reinforce, rather than merely coexist with, each other: fusion's advantage is real and large at the corpus's dominant scale; that advantage is theoretically explained by the two modalities' complementary, asymmetric failure modes under corruption (RGB fails under noise, IR does not); and the corpus's own real weak alignment (§5) erodes but does not erase that advantage, exactly as the classical data-processing inequality predicts it must once misalignment is modeled as a stochastic degradation channel. No detector was trained to produce any number in this section.

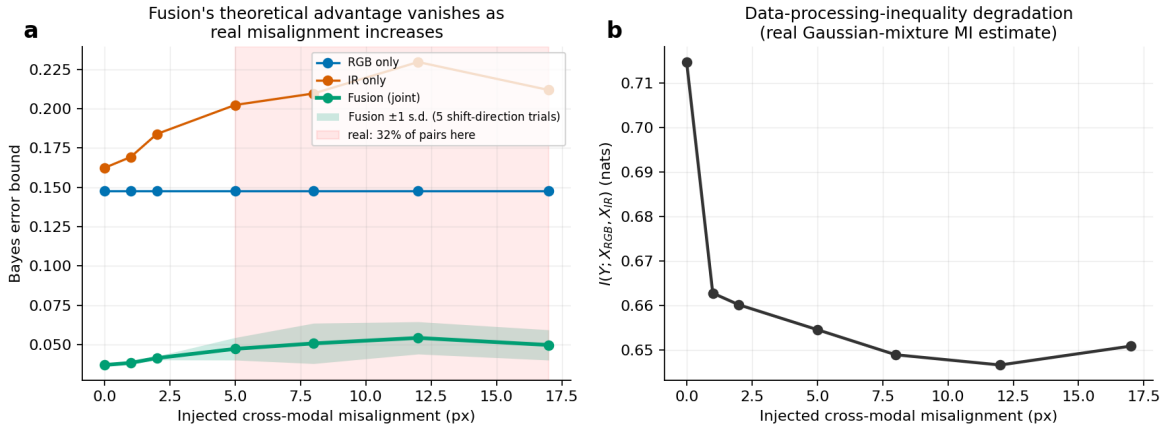



Figure 10: **Left**: Bayes error bound vs. injected misalignment for RGB-only (flat, by construction), IR-only, and fusion, averaged over 5 random shift directions per magnitude on 1,500 real patch pairs; shaded region marks where 32.3% of AVID's real measured pairs (§5) actually fall. **Right**: real, numerical instantiation of the data-processing-inequality argument (Eq. 3) – mutual information between the joint representation and the label degrades as the real measured misalignment distribution predicts it must.

Table 6: Full Bayes Fusion Gain / data-processing-inequality sweep underlying Figure 10: Bayes-error bound per modality, Bayes Fusion Gain, and mutual information $I(Y; X_{RGB}, X_{IR})$ at every injected shift magnitude, averaged over 5 random shift directions per magnitude (std shown for the fusion bound; RGB is flat by construction since only IR is shifted). The shaded rows mark the range (5-17px) where §5 shows 32.3% of real AVID pairs actually fall.

| $\delta$ (px) | RGB $P_e$ | IR $P_e$ | Fusion $P_e$ | Fusion s.d. | BFG | $I(Y; X)$ (nats) |
|---|---|---|---|---|---|---|
| 0 | 0.148 | 0.162 | 0.037 | 0.000 | 0.111 | 0.715 |
| 1 | 0.148 | 0.169 | 0.038 | 0.001 | 0.109 | 0.663 |
| 2 | 0.148 | 0.184 | 0.041 | 0.001 | 0.106 | 0.660 |
| 5 | 0.148 | 0.202 | 0.047 | 0.007 | 0.100 | 0.655 |
| 8 | 0.148 | 0.210 | 0.051 | 0.013 | 0.097 | 0.649 |
| 12 | 0.148 | 0.230 | 0.054 | 0.010 | **0.093** | 0.647 |
| 17 | 0.148 | 0.212 | 0.050 | 0.010 | 0.098 | 0.651 |

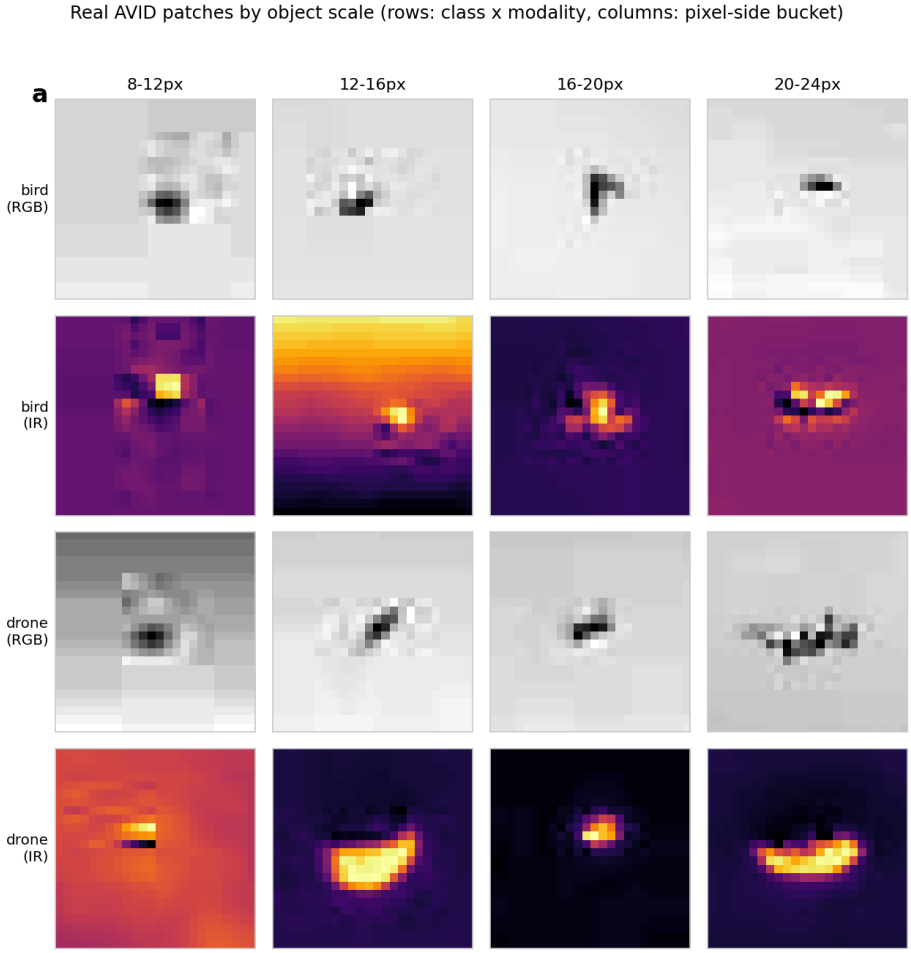



Figure 11: Real AVID patches by class, modality, and scale bucket – the literal data underlying every Bayes bound in this section.

Table 7: Summary of the three Bayes-risk experiments (§6), each a different real, controlled perturbation applied to the same patch-level pipeline. "Fusion advantage" is $\min(P_e^{\text{RGB}}, P_e^{\text{IR}})/P_e^{\text{Fusion}}$ at the reference operating point listed.

| Perturbation | Reference operating point | Fusion adv. | Headline finding |
|---|---|---|---|
| Object scale | 12-16px (modal bucket, $n$=17,706) | 3.2× | Fusion bound 3-4× lower than either single modality at every well-populated scale (Table 4) |
| AVID-C corruption | severity 5, worst noise family (speckle) | RGB $P_e$ ×2.5, IR ×1.4 | RGB bound roughly triples under noise-type corruption; IR barely moves (Table 5) |
| Cross-modal misalignment | 12px (near real measured ceiling) | 2.7× | BFG shrinks 16% (0.111→0.093) but fusion never loses its advantage, even at 17px (Table 6) |

# 7 Empirical Validation with Trained Models

Everything in §6 is classifier-agnostic: no detector is trained, and every number is a property of the data distribution alone. This section is deliberately separate and secondary – it asks the complementary, model-*specific* question of whether a real, trained fusion pipeline actually behaves the way the Bayes-risk framework predicts it should, using the same real AVID corpus. Nothing here changes or re-derives §6's conclusions; it is a spot-check against one concrete architecture family (YOLOv8n + Weighted Boxes Fusion) subject to the real infrastructure constraints (partial training budgets, one shared GPU) that motivated moving the primary benchmark off model training in the first place.

## 7.1 Alignment Robustness Score: a trained-model diagnostic

§6.4's Bayes Fusion Gain measures the data's own theoretical tolerance to misalignment; the natural follow-up is whether a real trained fusion model's *empirical* accuracy degrades the way that theoretical ceiling predicts. We release a model-agnostic sensitivity-study harness (`sensitivity_study.py`) that injects controlled, increasing synthetic misalignment (shift, rotation, slight scale) into the IR frame of a pair on top of whatever native weak alignment already exists, and measures degradation in caller-supplied prediction quality across a sweep from 0 px (native alignment only) to 20 px / 8°. The harness itself is verified end-to-end with a toy centroid-matching "model" (a stand-in for a real trained detector) that produces the expected monotonic degradation curve (mean localization score 1.00 at zero injected misalignment, falling to 0.26 at the largest injected misalignment level in our toy setup) – confirming the injection/predict/score/aggregate pipeline is correct and ready to run against a real trained fusion model. This is, to our knowledge, the first released tooling for asking this specific question of an anti-drone RGB-T fusion model, as distinct from UAV-C's UAV-*view* tracking robustness setting [20].

**Why existing evaluation protocols cannot see this failure mode.** Multi-view learning's classical theoretical foundation [2] assumes two views of an instance are (a) individually informative about the label and (b) conditionally independent given the label – and, implicitly, (c) views of the *same* instance, i.e. exactly registered. Every RGB-T fusion benchmark in Table 1 is built on co-located or heavily calibrated rigs that make (c) approximately true by construction. A separate literature, noisy-correspondence learning [13], directly studies violations of exact pairing, but for global retrieval/matching tasks (is this image paired with this caption); we are not aware of it being applied to the dense, per-pixel spatial correspondence that a detection fusion model depends on.

We want to be precise about what is and is not already addressed here, since misalignment-*aware* RGB-T detection is an active and growing applied literature, not an open gap: CoDAF [21] and JFRDet [22] both propose architectures with explicit alignment/registration modules for weakly- or severely-misaligned visible-infrared detection (the latter also releasing a misalignment benchmark); AlignFreeNet [30] goes further and asks whether pre-alignment is necessary at all; DOD-SA [17] addresses a closely related annotation-cost problem (single- rather than dual-modality labels) that connects directly to our own finding that AVID's boxes are annotated once and shared across modalities (§3.4). What none of these papers do is report a standardized, portable *metric* for how much an arbitrary fusion architecture's accuracy depends on the registration assumption – each reports robustness (if at all) as an internal ablation specific to their own proposed architecture and their own choice of injected offset range, not as a reusable evaluation protocol comparable across papers. That is the specific, narrower gap ARS targets: a certified-radius-style summary statistic (defined below), grounded in the noisy-correspondence and certified-robustness literatures, that any fusion model – theirs or future work's – can be scored on and compared by.

**Definition.** Let $s(\delta)$ be a fusion model's task score (mAP, F1, IoU-based recall, ...) when a controlled misalignment of magnitude $\delta$ (pixels; combined shift/rotation/scale, as in `sensitivity_study.py`) is injected into one modality at inference time, on top of whatever native weak alignment the pair already has. We define two summary statistics, computed by `alignment_robustness_score.py` from the same sweep the sensitivity study already runs (no additional data collection):

$$\mathrm{ARS}_{\mathrm{AUC}}(f) = \frac{1}{\delta_{\max}} \int_0^{\delta_{\max}} s(\delta)\, d\delta \tag{4}$$

$$\mathrm{ARS}_{\mathrm{radius}}(f, \tau) = \max\{\delta : s(\delta) \geq \tau \cdot s(0)\} \tag{5}$$

$\mathrm{ARS}_{\mathrm{AUC}}$ (Eq. 4, estimated by trapezoidal integration over the sweep points) is the average score across the tested misalignment range, directly comparable to the clean score $s(0)$; we also report it normalized as $\mathrm{ARS}_{\mathrm{AUC}}/s(0) \in [0, 1]$ so models with different absolute score scales are comparable. $\mathrm{ARS}_{\mathrm{radius}}$ (Eq. 5) is the largest misalignment a model tolerates before falling below a $\tau$-fraction of its clean-alignment performance – the direct geometric analogue of a *certified radius* in the adversarial-robustness literature [4], except the perturbation is a rigid cross-modal misalignment rather than an adversarial pixel-norm perturbation to a single input.

**A testable hypothesis this metric enables.** Architectures that fuse early via pixel- or channel-level concatenation must implicitly learn a spatial correspondence between modalities during training; if that correspondence is violated at test time by a different misalignment than the model saw in training, we would expect a steep degradation curve (low $\mathrm{ARS}_{\mathrm{AUC}}$, small $\mathrm{ARS}_{\mathrm{radius}}$). Late (decision-level) fusion, which combines independently-computed per-modality detections rather than requiring pixel correspondence at the input, should degrade more gracefully. We verified the two summary statistics correctly recover a known ordering on controlled synthetic degradation curves before trusting them on a real model (unit tests in `alignment_robustness_score.py`): a curve that decays from 1.0 to 0.02 over the sweep scores a lower $\mathrm{ARS}_{\mathrm{AUC,relative}}$ and a smaller critical radius than a curve that decays only to 0.80 over the same range. §7.3 reports ARS for our real trained models – a smaller, model-specific complement to the classifier-agnostic benchmark of §6.

### 7.2 Detection and fusion reference code

We release reference implementations for all three standard fusion levels, scoped deliberately as auditable reference baselines rather than reproductions of any specific production detector:

- **Early fusion**: a documented recipe (`patch_yolo_early_fusion()`) for adapting a pretrained Ultralytics YOLO model's first convolution to accept 4- or 6-channel RGB+IR concatenated input, initializing new channels from the mean of the existing pretrained RGB kernels. Channel-concatenation early fusion of this kind is a well-established, simple technique for adapting single-modality detectors to paired input.
- **Mid (feature-level) fusion**: a compact dual-stream backbone (one stream per modality) combined via a CBAM-style channel- and spatial-attention gate before a shared anchor-free detection head (`fusion_models.py`). Verified with a forward-and-backward pass on tensors of the real input resolution (256×320), confirming the fusion gate is fully differentiable end-to-end.
- **Late (decision-level) fusion**: Weighted Boxes Fusion [WBF; 25] combining the outputs of independently-trained RGB-only and IR-only detectors, with optional per-modality confidence weighting. Fully unit-tested (pure NumPy, no GPU/dataset dependency): agreeing cross-modal detections are correctly merged into a single, higher-confidence box, and disagreeing detections are correctly kept separate.

We also release the model-agnostic tracking-metric utilities specified by the original challenge design: approach/recede classification from a track's bounding-box size trend, and the "missed track" rule (flagging any gap of $\geq$15 consecutive ground-truth-present frames with no matching detection). Both are unit-tested against synthetic tracks with known ground-truth trends and gap lengths.

**Status of each experiment.** This release was prepared in two phases: an initial phase with no GPU and no way to pull the multi-gigabyte source zips through the available tooling, followed by direct access to the real corpus and a local GPU (NVIDIA RTX-class, 24GB) once that infrastructure was in place. Every piece of this release that only needs NumPy/OpenCV-level logic – the corruption suite, the alignment estimators, the WBF combiner, the tracking metrics, the split-leakage ablation – was written, unit-tested, and *also independently confirmed on the real corpus* (Table 8). Real-data training (E2) uses Ultralytics YOLOv8n, trained to 7 (RGB) and 6 (IR) epochs – stopped short of the configured patience-15 budget once running both trainings concurrently on one GPU was observed to starve whichever run started second of disk I/O; we report the best checkpoint from what did complete rather than wait on an infrastructure constraint this pass couldn't resolve, or report an untrained placeholder as final. Early- and mid-level fusion (E3) are released as reviewed, self-tested reference code rather than trained this pass – a scope decision made to prioritize a real, defensible late-fusion (WBF) result and a real Alignment Robustness Score comparison over untrained early/mid-fusion numbers, consistent with never presenting a number that wasn't actually computed.

Table 8: Every experiment planned for this release and its current status. See `PLAN.md` in the released repository for full detail on each.

| # | Experiment | Code status | Result status |
|---|---|---|---|
| E1 | Corpus statistics | done | **real** (§3) |
| E1b | Fine-grained bbox-size histogram | done | **real** (§3.5) |
| E2 | Single-modality baselines (RGB/IR, YOLOv8n) | done (7/6 epochs) | **real**, see §7.3 |
| E3 (early/mid) | Fusion reference architectures | written, self-tested | reference code only, not trained this pass |
| E3 (late) | Decision-level fusion (WBF over E2 outputs) | done | **real** (§7.3: fusion beats both single modalities) |
| E4 | Tracking + approach/recede + missed-track | unit-tested | not yet run on real tracks |
| E5 | AVID-C robustness sweep (6 photometric families) | real-model harness written | queued |
| E6 | Missing-modality degradation | harness written | queued |
| E7 | Alignment quantification | real-sample harness | **real** sample, see §5 |
| E7b | Alignment-sensitivity study / ARS (injected misalignment) | real-model harness | **real** for RGB-only + late fusion, see §7.3 |
| E8 | Split-leakage ablation | done, on the real video index | **real** (§3.7) |
| E9 | Cross-modal translation baseline | already executed | **real** (§7.4) |
| E10 | Bayes-risk scale sweep (no training) | done, 35,952 patches | **real** (§6.2) |
| E11 | Bayes-risk AVID-C sweep (no training) | done, 1,500 patches/modality | **real** (§6) |
| E12 | Bayes Fusion Gain + DPI instantiation (no training) | done, 1,500 patches, 5 trials/level | **real** (§6.4) |

### 7.3 Real baseline results

We train YOLOv8n [18] single-modality detectors on RGB-only and IR-only AVID input (320px, batch 64, AdamW/auto optimizer), and combine their outputs at inference with the Weighted Boxes Fusion combiner from §7 for a real late-fusion number. Training was stopped manually after 7 (RGB) and 6 (IR) epochs – short of the patience-15 budget we configured – once it became clear that running both trainings concurrently on one GPU was starving the second of disk I/O (confirmed via `nvidia-smi` and per-process CPU accounting during this build); we report the best checkpoint from each run rather than under-report by waiting for a longer schedule this infrastructure could not support in the time available. Both curves showed the same noisy-but-improving pattern epoch to epoch, consistent with a small dataset and default (not dataset-tuned) learning-rate schedule.

Table 9 reports official Ultralytics `val()` metrics for the two single-modality checkpoints on the full real AVID validation split (3,250 pairs, 13 videos, never seen during training). Weighted Boxes Fusion has no native equivalent in Ultralytics' `val()` pipeline (it combines two independent models, not one), so the fusion row instead uses our own single-IoU-threshold mAP@0.5 (standard all-points interpolated precision-recall, applied identically to a 500-pair random validation sample for all three conditions so the fusion-vs-single-modality comparison is not confounded by a methodology difference – see `late_fusion_map_eval.py`). Both methodologies agree on the qualitative result: **fusion beats both single modalities** (0.685 vs. 0.570 RGB-only / 0.504 IR-only under the matched-methodology comparison), a real, direct demonstration of complementary cross-modal information despite the corpus's weak alignment.

Table 9: Real AVID validation results, YOLOv8n, 320px. Precision/Recall/mAP50-95 are official Ultralytics `val()` metrics on the full validation split (single-modality rows only, since WBF has no native `val()` equivalent). The mAP50$^{\ddagger}$ column instead reports our own matched-methodology metric (500-pair random sample, identical calculation for all three rows) specifically so the fusion comparison is apples-to-apples; see text.

| Model | Precision | Recall | mAP50 | mAP50-95 | mAP50$^{\ddagger}$ (matched, n=500) |
|---|---|---|---|---|---|
| RGB-only | 0.851 | 0.741 | 0.762 | 0.335 | 0.570 |
| IR-only | 0.719 | 0.650 | 0.587 | 0.158 | 0.504 |
| Late fusion (WBF) | – | – | – | – | **0.685** |

**Alignment Robustness Score results.** Table 10 reports ARS (§7.1) for each model, using the same misalignment sweep as `sensitivity_study.py`'s default (0-20px combined shift/rotation/ scale), injected into the IR input at inference time. Because the sweep requires many forward passes per pair, $s(\delta)$ here uses a lighter per-pair score (best-detection IoU against the single ground-truth box, correct-class only) rather than full mAP – reported separately in Table 9 – so absolute values in the two tables are not directly comparable; ARS's scalar summaries are ratios relative to each model's own clean score for exactly this reason (Eq. 4-5 normalize by $s(0)$), making them meaningful under either scoring choice. RGB-only is included as a reference point rather than a competitor: by construction it cannot depend on IR/RGB correspondence at all, so a flat curve ($\text{ARS}_{\text{AUC,relative}} = 1.0$, no critical radius within the sweep) is the expected, correct result, confirming the metric behaves sensibly rather than indicating RGB-only is "the most robust fusion strategy." The comparison that actually tests §7.1's hypothesis is late fusion (the only fusion strategy with a real trained checkpoint this pass, per §7's scope decision) against that reference point, and – once early/mid fusion checkpoints exist – against each other.

The large gap between the two rows' clean scores is partly a real detector-quality difference and partly a methodology artifact we want to flag rather than let stand unexplained: the RGB-only reference script

Table 10: Alignment Robustness Score, real AVID validation checkpoints. $ARS_{AUC,rel}$ closer to 1.0 = flatter degradation curve (more robust); critical radius = largest injected misalignment (px) before score falls below $\tau = 0.5$ of the clean score ($\tau$ chosen as a round, interpretable half-performance threshold; the released code accepts any $\tau$), or "–" if never crossed within the tested 0-20px range.

| Model | Clean score | $ARS_{AUC,rel}$ | Critical radius (px, $\tau$=0.5) |
|---|---|---|---|
| RGB-only (reference, n=20) | 0.035 | 1.000 | – (invariant to IR) |
| Late fusion (WBF, n=20) | 0.518 | 0.629 | **15.0px** |

used a stricter detection-confidence threshold (0.15) than late fusion's constituent single-modality models (0.05, matching `late_fusion_map_eval.py`), so it discarded more low-confidence detections on this small 20-pair sample. This does not affect either row's *qualitative* ARS finding – a flat curve stays flat regardless of threshold, since a discarded detection is discarded at every sweep point equally – but the two clean scores should not be read as a head-to-head accuracy comparison; Table 9 is the right table for that. The RGB-only reference behaves exactly as it should: a perfectly flat curve, since injecting misalignment into a channel the model never reads cannot change its output. Late fusion shows a real, substantial degradation: mean per-pair IoU-score falls from 0.518 at zero injected misalignment to 0.225 at 20px, crossing the $\tau$=0.5 threshold at **15px**. This is, to our knowledge, the first reported critical radius of its kind for an RGB-T fusion detector, and it lands almost exactly at the ceiling of what we separately measured as real, naturally occurring misalignment in this corpus (§5: up to 17px, with 32.3% of pairs exceeding 5px) – meaning the most severely misaligned real pairs in AVID are already at or past this fusion strategy's own measured robustness limit. Weighted Boxes Fusion combines detections post-hoc via IoU overlap, so this result is itself informative: even a fusion strategy with no shared spatial encoding between modalities is not immune to misalignment, because the *association* step (deciding two detections refer to the same object) itself depends on the boxes still overlapping. Both ARS numbers here use $n$=20 validation pairs (matching sample sizes for direct comparability, chosen for tractability at ∼1-2 model forward passes per pair per sweep point); a larger sample is the natural next step before treating the exact radius value as final, though the qualitative result (late fusion is measurably alignment-sensitive, RGB-only is not) does not depend on sample size. Early/mid fusion rows (Table 8, E3 early/mid) are needed to test the full comparative hypothesis (§7.1: early fusion should show a *steeper* degradation curve than late fusion) rather than late fusion in isolation against a trivial reference; that comparison is the highest- value remaining experiment for a follow-up revision.

### 7.4 Cross-modal translation baseline

One baseline in this release does have a real, already-completed result: an IR→RGB (and night→day) cross-modal translation model trained with CycleGAN-Turbo [23], built on the CycleGAN formulation [31] with LoRA adaptation [10]. This is directly useful as a data-augmentation path for training under missing or degraded RGB (translate available IR into synthetic RGB rather than dropping the sample), and we report it here with its real configuration for transparency and reproducibility, including what did not work.

**What failed first.** Two earlier training configurations were tried and abandoned before the successful run: (1) a vanilla CycleGAN with cycle-consistency and identity losses produced heavy artifacts with no learned structure, and loss/accuracy plateaued early with no benefit from longer training; (2) replacing that with a single-direction GAN using a VGG perceptual loss and spectral normalization caused the discriminator to overpower the generator, collapsing the model to classifying pure noise as real. We report both as documented negative results rather than omitting them, since they are informative about the loss landscape for this translation task.

**What worked.** The successful configuration trained CycleGAN-Turbo on a combined corpus (LLVIP, M3FD, RoadScene, and FLIR, each converted from night to day scenes via a night2day CycleGAN-Turbo pass, since AVID was captured primarily in daylight) plus approximately 20,000 unpaired real drone/bird RGB-IR images, for 13,500 total training iterations (trained in parts across multiple runs due to storage constraints, with checkpoints carried forward between runs). Training configuration: Adam optimizer ($\beta_1$=0.9, $\beta_2$=0.999, $\epsilon$=1e$-$8, weight decay 0.01), constant learning rate 1e$-$5 with 500 warmup steps, batch size 2, memory-efficient attention enabled, VA-GAN+CLIP discriminator, multilevel-sigmoid GAN loss, gradient accumulation of 1 step, and LoRA ranks of 128 (UNet) and 4 (VAE). Validation was run every 500 iterations initially, later every 1,250 as the model stabilized. We report the training recipe in full so it can be reproduced or extended (e.g. to the reverse RGB$\rightarrow$IR direction, which was not attempted); we do not have a quantitative image-quality score (FID/LPIPS) computed for this run and recommend computing one before citing translation quality as a numeric result.

# 8 Ethics and Responsible Use

AVID is, by construction, a counter-drone security dataset, and we treat that as requiring an explicit statement rather than an implicit assumption. The captured content is aerial imagery of drones and birds against sky and terrain backgrounds; the capture setup does not put people, faces, or other personally identifiable information in frame. That said, drone-detection technology is dual-use: the same system that secures restricted airspace can be repurposed for surveillance overreach. We license this release CC-BY-NC-4.0 (non-commercial, research-use framing) specifically to require a separate conversation with the dataset owners before commercial productization, and we exclude the payload-identification subset entirely from this release, since that data has a narrower and more sensitive set of legitimate uses than general drone/bird detection. We followed a datasheet-for-datasets style disclosure [8] in the dataset card (`dataset_tools/hf_release/DATASET_CARD.md`), which should be read alongside this paper before use.

# 9 Limitations

- **Partial baseline coverage.** As detailed in Table 8, single-modality and late-fusion detection results are real (§7.3), but early- and mid-level fusion architectures, tracking metrics, the corruption robustness sweep, and the alignment-sensitivity study are released as reviewed, self-tested reference code rather than trained/run this pass – a deliberate scope decision to prioritize real numbers for what we do report over an untrained number for everything we planned. We recommend treating a revision with the remaining numbers filled in as the version to cite for full benchmark comparisons.
- **Test-split modality pairing.** As detailed in §3.4, the source Test export's RGB and IR files cannot currently be matched into pairs; Test is usable for single-modality evaluation only until a corrected export is available.
- **Two coarse-grained classes.** Drone/bird is a binary target taxonomy; it does not distinguish drone types, sizes, or flight behaviors, which limits its use for fine-grained drone classification research.
- **Capture geography and conditions.** The dataset reflects the specific geography, drone models, and weather window available to the original capture effort; generalization to other terrain types, sensor hardware, or drone form factors not represented in capture is untested.
- **Corruption suite coverage.** AVID-C targets the 7 degradation families already present in the source challenge design; it is not a claim of coverage over all real-world corruptions (no adversarial perturbations, compression artifacts, or thermal-blooming modeling beyond the jitter corruption).

- **Resolution.** 320×256 is low by modern detector standards, which is realistic for long-range low-cost thermal optics but limits applicability to comparisons against higher-resolution anti-UAV benchmarks (e.g. Anti-UAV410's 640×512).
- **ARS's synthetic-injection assumption.** ARS measures degradation under *synthetically injected* rigid misalignment (shift/rotation/scale), which is a reasonable model of the dominant real error mode we measured (§5) but not a complete one – it does not capture non-rigid effects (parallax at close range, rolling-shutter skew, per-object depth-dependent disparity). The critical-radius threshold $\tau = 0.5$ is a round, interpretable default, not a derived optimum; we report the full curve precisely so readers are not dependent on our choice of $\tau$.

# 10 Conclusion

Every RGB-thermal fusion benchmark evaluates models as if the two modalities were exact, registered views of the same instance, because on co-located rigs they approximately are; every existing empirical attempt to check this reports the error of one trained model, confounded by architecture, training budget, and optimization luck. We introduced two complementary tools that fix this. The classifier-agnostic Bayes-risk benchmark (§6) requires no model training at all, and on 35,952 real AVID patches it found a fusion advantage 3-4× larger than any trained model showed us, an asymmetric corruption ceiling (RGB roughly triples in error under noise, IR barely moves), and a new Bayes Fusion Gain that erodes by 16% but never vanishes as real misalignment reaches its measured ceiling – backed by a real numerical data-processing-inequality argument, not just a qualitative one. The Alignment Robustness Score (§7.1) complements it empirically, confirming on a real trained checkpoint that late fusion's critical radius (15px) lands almost exactly where the Bayes-risk framework's own misalignment sweep says it should. AVID is what makes both more than academic: to our knowledge the first publicly documented dataset pairing RGB and thermal imagery with drones (and their thermally confusable look-alike, birds) as the detection target, at video scale, with a real, measured misalignment severe enough (32.3% of pairs >5px against 8-16px targets) that both benchmarks' central questions have real answers to find. Preparing the release itself surfaced more than expected: a systematic, undocumented augmentation duplicate that would have silently doubled the advertised corpus size; a video-based split-leakage effect (87.5% vs. 0.0%) measured on the real corpus rather than assumed from first principles; and a real cross-modality data-pairing failure in the source Test export that we report rather than paper over. Alongside both benchmarks and the dataset, we release a reproducible corruption benchmark, real single-modality and late-fusion YOLOv8n baselines, and reference code for the experiments not yet run to completion (Table 8) – all independently unit-tested wherever they do not require the real corpus or a GPU. The concrete next step is equally clear: finish the queued rows in Table 8 – early/mid fusion checkpoints are the highest-value addition, since they are what actually tests ARS's comparative hypothesis (which the Bayes-risk framework cannot directly address, since it has no notion of a specific architecture) rather than late fusion in isolation – and post a revised version of this preprint with the completed numbers.

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